Product Requirements Document (PRD)

Knowledge Management System & Website Upgrade

Plateau State Peacebuilding Agency (PSPBA)

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1. Executive Summary

This PRD defines the requirements for a comprehensive Knowledge Management System (KMS) and Website Upgrade for the Plateau State Peacebuilding Agency (PSPBA). The project addresses critical gaps in systematic knowledge capture, storage, and dissemination identified during the assessment phase. The solution will transform PSPBA into an evidence-based learning hub through a centralized digital repository, modern responsive website, conflict data visualization tools, and streamlined application management for volunteers, interns, and consultants.

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2. Problem Statement

PSPBA currently faces significant challenges in managing organizational knowledge and digital presence:

Knowledge Management Issues:

- Fragmented data storage across decentralized locations causing duplication and inconsistency
- Inconsistent documentation practices without standardized templates or procedures
- Knowledge loss during staff transitions due to lack of systematic archiving
- Weak cross-unit learning and information sharing mechanisms

Digital Infrastructure Issues:

- Outdated website (plateaupeacebuilding.org) with poor mobile responsiveness
- Limited content management capabilities for publishing reports and multimedia
- No centralized digital repository for document storage and retrieval
- Poor email synchronization affecting communication efficiency
- No visualization tools for conflict trend analysis across 17 LGAs

3. User Personas

3.1 Internal Users

PSPBA Staff Member

- Role: Programme Officers, MERL staff, Communications team
- Needs: Upload documents, search institutional knowledge, track conflict data, publish website content
- Pain Points: Cannot find historical documents, inconsistent access to information, manual processes for everything

PSPBA Management

- Role: Director-General, Department Heads
- Needs: Generate reports, monitor organizational activities, track KM contributions, view analytics
- Pain Points: No visibility into knowledge usage, difficulty in evidence-based decision making

ICT Unit Staff

- Role: Technical administrators
- Needs: Manage user access, maintain systems, handle backups, troubleshoot issues
- Pain Points: No centralized system to administer, inadequate tools

3.2 External Users

Public Visitor

- Role: Researchers, media, citizens, partner organizations
- Needs: Access public reports, view conflict trends, learn about PSPBA activities
- Pain Points: Difficult navigation, outdated information, no interactive features

Applicant (Volunteer/Intern/Consultant)

- Role: Individuals seeking engagement with PSPBA
- Needs: Submit applications online, receive confirmation, track status
- Pain Points: Currently no online application process

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4. Functional Requirements (FRs)

Epic 1: Knowledge Management System

FR-1.1: Centralized Digital Repository

- FR-1.1.1: Users SHALL upload documents to cloud-based storage with local backup
- FR-1.1.2: System SHALL categorize documents by project, thematic area, date, and document type
- FR-1.1.3: System SHALL support metadata tagging for advanced search and retrieval
- FR-1.1.4: System SHALL maintain version control with complete edit history

FR-1.2: User Dashboards

- FR-1.2.1: System SHALL provide department-specific dashboards (MERL, Strategic Communications, ICT, Programme)
- FR-1.2.2: Users SHALL upload documents through intuitive interface with automatic metadata capture
- FR-1.2.3: System SHALL implement role-based access controls for data security

FR-1.3: Advanced Search

- FR-1.3.1: Users SHALL search by keywords, date range, project name, thematic area, author, or document type
- FR-1.3.2: System SHALL provide full-text search within document contents
- FR-1.3.3: System SHALL offer filters, facets, and search suggestions

FR-1.4: Analytics & Reporting

- FR-1.4.1: System SHALL track document access (who, what, when, frequency)
- FR-1.4.2: System SHALL identify documentation gaps requiring additional knowledge capture
- FR-1.4.3: Users SHALL generate custom reports on KM activities and trends

Epic 2: Conflict & Peace Data Management

FR-2.1: Incident Database

- FR-2.1.1: Users SHALL record conflict incidents with standardized fields
- FR-2.1.2: System SHALL maintain database of peacebuilding actors with contact details and focus areas

FR-2.2: Geographic Visualization

- FR-2.2.1: System SHALL display interactive map of Plateau State showing all 17 LGAs
- FR-2.2.2: Map SHALL use color-coded legend for conflict intensity and peacebuilding interventions
- FR-2.2.3: Map SHALL link to database records for real-time data visualization

FR-2.3: Trend Analysis

- FR-2.3.1: Users SHALL analyze conflict patterns, triggers, and intervention effectiveness

Epic 3: Website Redesign

FR-3.1: Domain & Design

- FR-3.1.1: System SHALL migrate from plateaupeacebuilding.org to pspba.org with proper 301 redirects
- FR-3.1.2: Website SHALL implement mobile-first responsive design
- FR-3.1.3: Website SHALL load quickly even on slow connections

FR-3.2: Content Management

- FR-3.2.1: Staff SHALL update content without technical expertise via user-friendly admin interface
- FR-3.2.2: System SHALL enable direct publishing from KMS to website
- FR-3.2.3: System SHALL implement content approval workflow before publication

FR-3.3: Multimedia Capabilities

- FR-3.3.1: Website SHALL support photo galleries, embedded videos, and success stories
- FR-3.3.2: Website SHALL provide public document repository for reports, briefs, and publications

FR-3.4: Interactive Features

- FR-3.4.1: Website SHALL include news/blog section, events calendar, and contact forms
- FR-3.4.2: Visitors SHALL subscribe to newsletter updates
- FR-3.4.3: Website SHALL display interactive conflict trend map

Epic 4: Online Application Portal

FR-4.1: Application Forms

- FR-4.1.1: Volunteers SHALL submit basic form (name, contact, area of interest, motivation, optional CV)
- FR-4.1.2: Interns SHALL submit form (name, contact, institution, preferred department, CV upload)
- FR-4.1.3: Consultants SHALL submit form (name, contact, expertise area, CV/proposal upload)

FR-4.2: Notifications

- FR-4.2.1: System SHALL send automatic confirmation email to applicants upon submission
- FR-4.2.2: System SHALL notify PSPBA staff of new applications

Epic 5: Email System Upgrade

FR-5.1: Professional Email

- FR-5.1.1: System SHALL migrate to Microsoft 365 or Google Workspace with @pspba.org addresses
- FR-5.1.2: System SHALL provide seamless Outlook synchronization (desktop and mobile)
- FR-5.1.3: System SHALL enable calendar sharing and team collaboration features

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5. Non-Functional Requirements (NFRs)

5.1 Performance

ID	Requirement	Target
NFR-P1	Page load time	< 3 seconds on 3G connection
NFR-P2	Concurrent users	Support 100+ simultaneous users
NFR-P3	Document storage	Unlimited with expansion capability
NFR-P4	System uptime	99.9% availability

5.2 Security

ID	Requirement	Target
NFR-S1	Data transmission	SSL/TLS encryption for all traffic
NFR-S2	Access control	Role-based access control (RBAC)
NFR-S3	Data encryption	Encryption at rest and in transit
NFR-S4	Security audits	Regular vulnerability assessments
NFR-S5	DDoS protection	Active protection against cyberattacks

5.3 Reliability

ID	Requirement	Target
NFR-R1	Automated backups	Daily with 30-day retention

ID	Requirement	Target
NFR-R2	Disaster recovery	4-hour recovery time objective
NFR-R3	Data residency	Nigerian data residency compliance

5.4 Usability

ID	Requirement	Target
NFR-U1	Device compatibility	Desktop, tablet, and mobile access
NFR-U2	Offline access	Critical functions available offline
NFR-U3	Training time	Staff proficient within 2-hour training

5.5 Maintainability

ID	Requirement	Target
NFR-M1	Content updates	Non-technical staff can update content via admin interface
NFR-M2	Documentation	Complete operational manual provided
NFR-M3	Support	Post-launch support agreement

6. Success Metrics

Metric	Baseline	Target	Measurement Method
Document upload rate	0	50+ documents/month	KMS analytics
Search success rate	N/A	>80% finding desired document	User surveys
Website traffic	Current baseline	50% increase	Google Analytics
Mobile responsiveness	Not mobile-friendly	100% responsive	Testing tools
Staff KMS adoption	0%	>80% active users	Login tracking
Application processing time	Manual/undefined	<48 hour acknowledgment	System logs

Metric	Baseline	Target	Measurement Method
Cross-unit document sharing	Minimal	>30% shared documents	KMS analytics

7. Technical Architecture Summary

7.1 System Components

Component	Specification
Platform Type	Web-based, cloud-hosted with local backup
Hosting	Secure cloud with Nigerian data residency
Frontend	React.js with responsive design (mobile-first)
Backend	Node.js / Express.js API
Database	PostgreSQL / MySQL with automated backups

7.2 Integration Points

- KMS ↔ Website (content publishing)
- Database ↔ Interactive Map (conflict visualization)
- Application Forms ↔ Email Notifications
- Cloud Storage ↔ Local Backup System

8. Risks and Assumptions

8.1 Risks

Risk	Impact	Probability	Mitigation
Staff resistance to new systems	Low adoption	Medium	Comprehensive change management, leadership support
Technical failures/downtime	Operations disruption	Low	Reliable hosting, regular backups, disaster recovery

Risk	Impact	Probability	Mitigation
Insufficient training time	Poor system usage	Medium	Training manuals, video tutorials, post-training support
Data security breaches	Sensitive data compromise	Low	Strong encryption, RBAC, regular security audits
Power/internet connectivity issues	Limited system access	High	Offline capability, local backup, mobile data alternatives
Budget constraints for maintenance	System deterioration	Medium	Low-cost design, budget advocacy, partner support

8.2 Assumptions

- PSPBA leadership will actively champion system adoption
- Staff will participate in training programs
- Reliable internet connectivity available at PSPBA offices
- Budget allocated for ongoing hosting and maintenance costs
- Existing data can be migrated to new systems
- Domain pspba.org is available for registration

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9. Implementation Timeline

Phase	Activities	Timeline
Phase 1	Foundation & Setup	Nov 21, 2025
Phase 2	Core System Development	Nov 22 - Dec 31, 2025
Phase 3	Integration & Testing	Jan 6 - 27, 2026
Phase 4	Training & Capacity Building	Jan 6 - 27, 2026
Phase 5	Launch & Validation	Jan 6 - 27, 2026
Phase 6	Stabilization & Handover	Jan 6 - 27, 2026

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10. Out of Scope

The following items are explicitly excluded from this project:

- Development of mobile applications (native iOS/Android)
- Integration with external government databases
- Advanced AI/ML features for conflict prediction
- Multi-language support beyond English
- Social media account management
- Hardware procurement (computers, servers)

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11. Dependencies

- Availability of @pspba.org domain
- PSPBA staff availability for requirement validation and training
- Access to existing documents and data for migration
- Cloud hosting provider selection and setup
- SSL certificate procurement

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12. Acceptance Criteria

The project will be considered complete when:

1. KMS is operational with all FRs implemented and tested
2. Website is live on pspba.org with responsive design verified
3. Application portal accepts and processes submissions
4. Email system migrated and synchronized across devices
5. Interactive map displays conflict data for all 17 LGAs
6. Staff training completed with documented proficiency
7. Operational manual and documentation delivered
8. 30-day post-launch support completed without critical issues

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Appendices

Appendix A: Glossary

Term	Definition
KMS	Knowledge Management System
LGA	Local Government Area
MERL	Monitoring, Evaluation, Research and Learning
PSPBA	Plateau State Peacebuilding Agency
RBAC	Role-Based Access Control
SPRiNG	Strengthening Peace and Resilience in Nigeria

Appendix B: Document References

- Milestone 1: Assessment & Design Report (November 2025)
- PSPBA Road Map to Peace Initiative Documentation
- ICT Infrastructure Audit Findings

Document prepared following BMAD Method PRD framework